

Part I

## Recitation Questions

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### Problem 1

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February 21, 2026

*Problem 1*

**Problem 1** Determine whether the following statements are true or false and give either an explanation or a counterexample.

- (a) The function  $f(x) = \sqrt{x}$  has a local maximum on the interval  $[0, 1]$ .
- (b) If  $f'(2) = 0$ , then  $x = 2$  is either a local maximum or local minimum of  $f$ .

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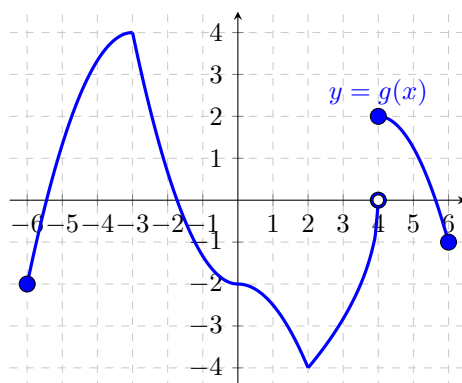
## Problem 2

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February 21, 2026

Problem 2

**Problem 2** The entire graph of a function  $g$  is given below.



Based on the graph of  $g$ , answer the questions below.

- List the  $x$ -coordinates of all critical points of  $g$ .
- List the  $x$ -coordinates of all critical points of  $g$  where  $g'(x) = 0$ .
- List the  $x$ -coordinates of all critical points of  $g$  where  $g'(x)$  is **undefined**.
- List the  $x$ -coordinates of all local maximums of  $g$ .
- List the  $x$ -coordinates of all local minimums of  $g$ .
- List all intervals where  $g$  is both decreasing AND concave down.
- List all intervals where  $g$  is both decreasing AND concave up.
- List all intervals where  $g$  is both increasing AND concave down.
- List all intervals where  $g$  is both increasing AND concave up.
- List the  $x$ -coordinates of all inflection points of  $g$ .

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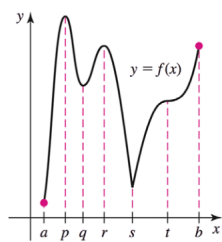
# Problem 3

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Problem 3

**Problem 3**  
*below, determine  
that point.*



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# Problem 4

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*Problem 4*

**Problem 4** *Locate the critical points and use the second derivative test to determine whether they correspond to local maxima or local minima. **EX-PLAIN.***

$f(x) = (x + c)^4$  where  $c$  is a positive constant

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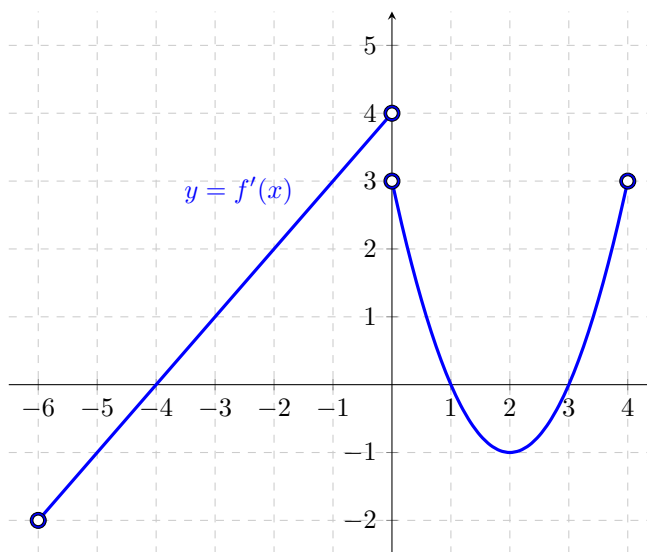
# Problem 5

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Problem 5

**Problem 5** Let  $f$  be a function that is continuous on its domain  $(-6, 4)$ . The graph of  $f'$ , the derivative of  $f$ , is given below.



Based on the graph of  $f'$ , answer the question below.

- List x-coordinates of all critical points of  $f$ .
- List x-coordinates of all critical points of  $f$  where  $f'(x) = 0$ .
- List x-coordinates of all critical points of  $f$  where  $f'(x)$  is **undefined**.
- List x-coordinates of all local minimums of  $f$ .
- List x-coordinates of all local maximums of  $f$ .
- List all intervals where  $f$  is decreasing and concave down.
- List all intervals where  $f$  is decreasing and concave up.
- List all intervals where  $f$  is increasing and concave down.
- List all intervals where  $f$  is increasing and concave up.
- List x-coordinates of all inflection points of  $f$ .

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# Problem 6

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Problem 6

**Problem 6** Find the critical points of  $f$  on the given interval. Determine whether the function  $f$  has a local maximum, local minimum or no local extremum at each critical point. **EXPLAIN**.

(a)  $f(x) = x\sqrt{2-x^2}$  on  $(-\sqrt{2}, \sqrt{2})$ .

(b)  $f(x) = x^3e^{-x}$  on  $(-1, 5)$ .

(c)  $f(x) = x \ln \left( \frac{x}{5} \right)$  on  $(0, 5)$ .

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# Problem 7

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Problem 7

**Problem 7** Let  $f(x) = \frac{1}{1+x^2}$ . Find the following for  $f$ :

- (a)  $f'$  and  $f''$
- (b) Critical points
- (c) Local extrema (and check your answers with both the first and second derivative tests)
- (d) Inflection points. **EXPLAIN**.

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# Problem 8

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*Problem 8*

**Problem 8** *Sketch a possible graph of a function  $f$  that has the following properties:*

- (a)  *$f$  is defined on the interval  $(0, 6)$ .*
- (b)  *$f$  has no local maximums.*
- (c)  *$f$  has exactly two local minimums.*

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# Problem 9

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*Problem 9*

**Problem 9** Consider the parabola  $f(x) = ax^2 + bx + c$  where  $a, b, c$  are constants. For what values of  $a, b, c$  is  $f$  concave up? For what values of  $a, b, c$  is  $f$  concave down?